

UNIVERSITY OF WAH(UOW)
DEPARTMENT OF COMPUTER
SCIENCE



Final Lab Sessional

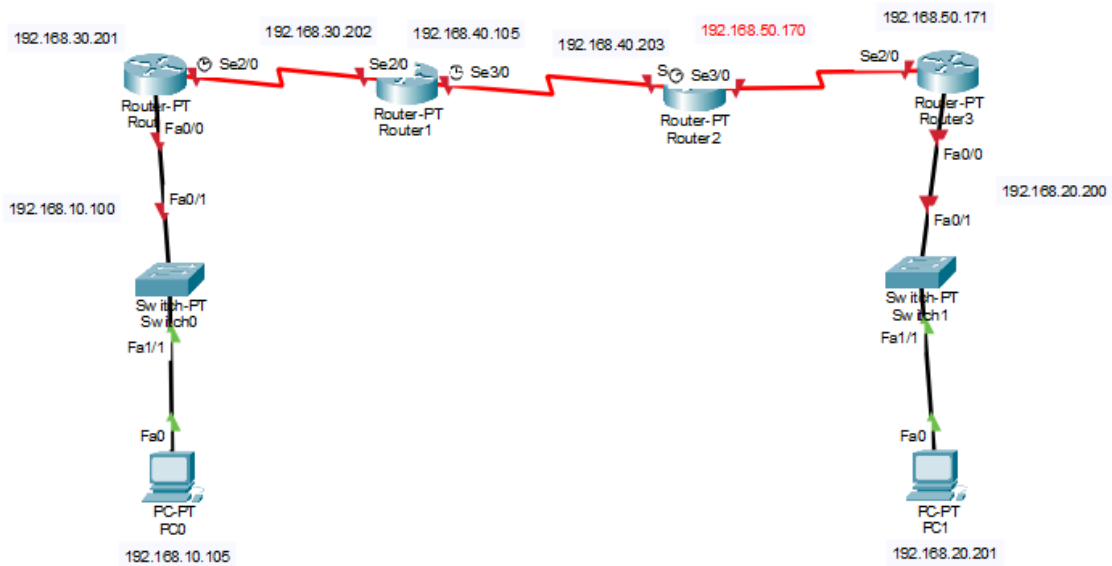
COURSE: Computer Networks Lab_241L

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SUBMITTED BY: Muhammad Talha_CS.3ndB

Registration NO: UW-24-CS-BS-101

Task 2



Router 1

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 10
Router(config-router)#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.

Router(config-router)#neighbor 192.168.30.202 remote-as 20
Router(config-router)#network 192.168.10.0 mask 255.255.255.0
Router(config-router)#network 192.168.30.0 mask 255.255.255.0
Router(config-router)#
```

Router 2

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 20
Router(config-router)#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.

Router(config-router)#neighbor 192.168.30.201 remote-as 10
Router(config-router)#neighbor 192.168.30.203 remote-as 30
Router(config-router)#network 192.168.30.0 mask 255.255.255.0
Router(config-router)#network 192.168.40.0 mask 255.255.255.0
Router(config-router)#
```

Router 3

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 30
Router(config-router)#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.

Router(config-router)#
Router(config-router)#
Router(config-router)#neighbor 192.168.40.202 remote-as 20
Router(config-router)#neighbor 192.168.50.171 remote-as 40
Router(config-router)#
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
|
```

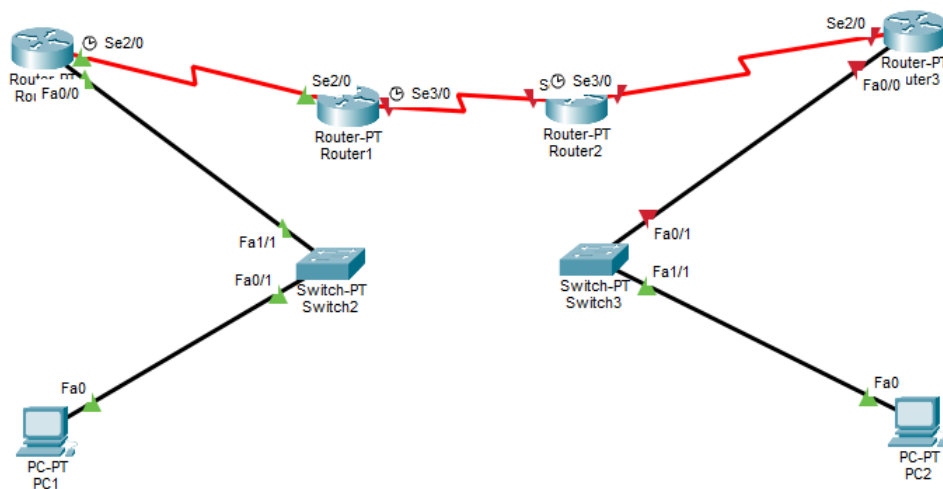
Router 4

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 40
Router(config-router)#
Router(config-router)#neighbor 192.168.50.170 remote-as 30
Router(config-router)#
Router(config-router)#network 192.168.20.0 mask 255.255.255.0
Router(config-router)#end
Router#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.
```

Check Messaging

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Router0	Router1	ICMP		0.000	N	0	(e...	(delete)
	Successful	Router1	Router2	ICMP		0.000	N	1	(e...	(delete)
	Successful	Router2	Router3	ICMP		0.000	N	2	(e...	(delete)

Task 3



Every Router

```
Router(config)#
Router(config)#
Router(config)#router eigrp 1
Router(config-router)#
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.30.0 0.0.0.255
Router(config-router)#network 192.168.40.0 0.0.0.3
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Sending

	Successful	PC3	PC5	ICMP		0.000	N	1	(e...	(delete)
	Successful	PC3	PC4	ICMP		0.000	N	2	(e...	(delete)
	Successful	PC5	PC3	ICMP		0.000	N	3	(e...	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	4	(e...	(delete)

Task 1

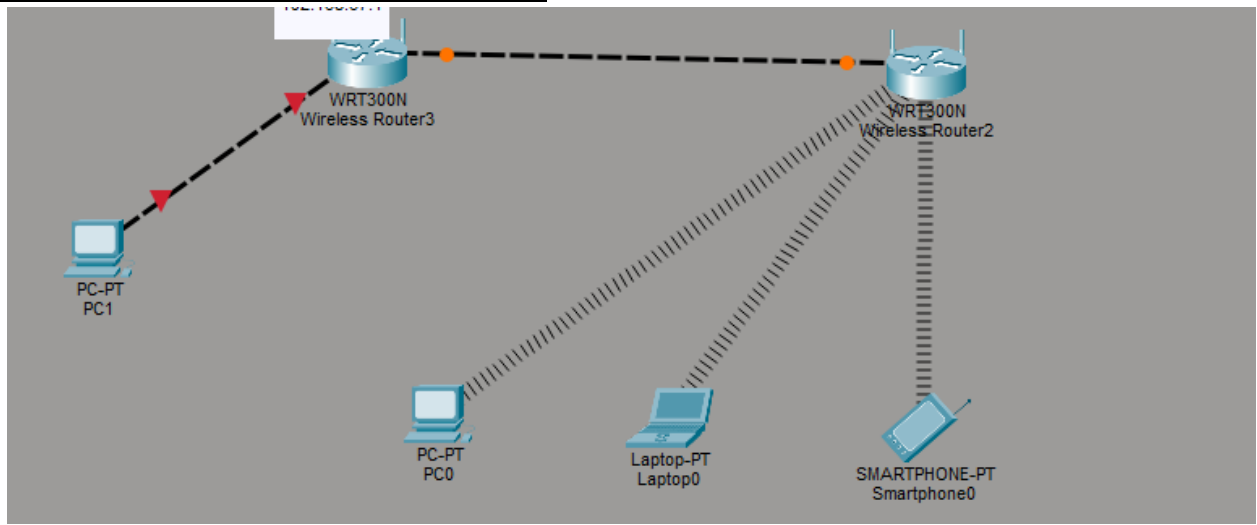
Task 01:

A family home has multiple smartphones and laptops that need secure wireless access while a desktop computer is connected via cable; DESIGN a wireless network using 2 wireless router, 3 wireless devices, and 1 wired PC, configure wireless security and IP addressing, and confirm that all devices can communicate successfully within the network

Solution:

Step 01:

Add two wireless routers in the network and also add one switch and also add three devices three are wireless and one pcs are connected via cable:



Step 02:

Give Ip Addresses according to the instructions:

Router 01:

IP Configuration	
IPv4 Address	192.168.67.1
Subnet Mask	255.255.255.0

PC 01 connected via cable

☒ DHCP	☐ Static
IPv4 Address	192.168.67.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.67.1
DNS Server	0.0.0.0
IPv6 Configuration	

Wireless pcs and laptops and msart phone:

☒ DHCP	☒ Static
IPv4 Address	192.168.152.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.152.1
DNS Server	0.0.0.0

☒ DHCP	☒ Static
IPv4 Address	192.168.152.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.152.1
DNS Server	0.0.0.0

☒ DHCP	☒ Static
IPv4 Address	192.168.152.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.152.1
DNS Server	0.0.0.0

IPv6 Configuration

Step 03: Send messages:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	D
	Successful	PC0	Laptop0	ICMP		0.000	N	0	(e... (d	
	Successful	Laptop0	Smartphone0	ICMP		0.000	N	1	(e... (d	
	Successful	Smart...	Laptop0	ICMP		0.000	N	2	(e... (d	
	Successful	Laptop0	PC0	ICMP		0.000	N	3	(e... (d	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	D
	Successful	Laptop0	Wireless R...	ICMP		0.000	N	5	(e... (d	
	Successful	PC0	Wireless R...	ICMP		0.000	N	6	(e... (d	
	Successful	Smart...	Wireless R...	ICMP		0.000	N	7	(e... (d	